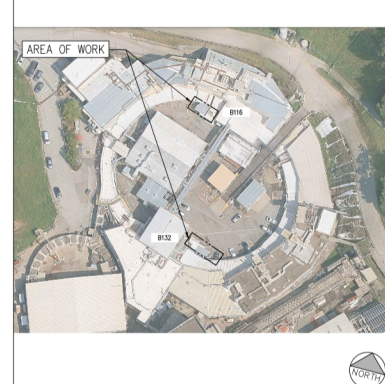


## B116 AND B132 HVAC DESIGN

## AREA OF WORK

[illegible]

|                                                                                                                                                           |                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>SLAC BUILDING INSPECTION OFFICE</b>                                                                                                                    |                                                                     |
| <b>Construction Authorization (IFC)</b>                                                                                                                   |                                                                     |
| BY: onslater                                                                                                                                              |                                                                     |
| DATE: 1/23/2026                                                                                                                                           | PRSR: 25-254                                                        |
| Revised Authorization                                                                                                                                     | <input checked="" type="checkbox"/> No <input type="checkbox"/> Yes |
| These plans have been reviewed for compliance with applicable building codes and standards, and IBC program requirements.                                 |                                                                     |
| This authorization of these plans shall not be construed to be a permit or an approval for violation of any applicable codes, standards, or IBC programs. |                                                                     |
| <b>THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.</b>                                                                        |                                                                     |

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

THESE PLANS SHALL BE KEPT ON THE JOB SITE  
FOR ALL CONSTRUCTION INSPECTIONS.

**vektor** Engineering & Consulting Services, Inc.  
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|                 |                      |               |
|-----------------|----------------------|---------------|
| SCALE: AS NOTED | DO NOT SCALE DRAWING | S5010.2025.01 |
|-----------------|----------------------|---------------|

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| G0.01 |
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|  |                           |
|--|---------------------------|
|  | B116 AND B132             |
|  | B116 AND B132 HVAC DESIGN |
|  | GENERAL                   |
|  | COVER SHEET               |

|                     |           |              |      |
|---------------------|-----------|--------------|------|
| ENGR HORLBY, DEBORA | DATE      | APPROVALS    | DATE |
| DLS FERNANDEZ, HUGO | 12-8-2025 | BO-HOFF, AND | 12-  |

|                  |                                  |               |   |
|------------------|----------------------------------|---------------|---|
| DATE<br>10-02-04 | DRAWING NUMBER<br>R0116 C00 0001 | REVISION<br>D | E |
|------------------|----------------------------------|---------------|---|

|                        |   |
|------------------------|---|
| B0116 - G00 - 0001 - D | E |
|------------------------|---|

| GENERAL NOTES                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | ABBREVIATIONS   |                                                |
|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------------------------------------|
| 1.                                                                                                        | ALL WORK SHALL CONFORM TO THE 2022 CALIFORNIA MECHANICAL CODE.                                                                                                                                                                                                                                                                                                                                                                                                             | ABREV           | DESCRIPTION                                    |
| 2.                                                                                                        | THE SUBMITTAL OF A BID PROPRIAL SHALL BE CONSIDERED CONCLUSIVE EVIDENCE THAT THE CONTRACTOR IS THOROUGHLY FAMILIAR WITH THE INTENT OF THE CONTRACT DOCUMENTS. THE CONTRACTOR PRIOR TO BIDDING, SHALL CHECK SITE CONDITIONS, INSTALLATIONS AND LISTINGS RELATED TO EXISTING UTILITIES, INCLUDING BUT NOT LIMITED TO, ELECTRICAL AND MATERIAL REQUIRED FOR COMPLETION OF THE SYSTEM.                                                                                         | BTH             | BOTH THERMAL INLET, BOTH THERMAL UNIT PER HOUR |
| 3.                                                                                                        | CONTRACTOR SHALL BE RESPONSIBLE FOR HAVING VISITED THE SITE AND SATISFIED HIMSELF THAT THE CONTRACTOR'S UNDERSTANDING OF THE WORK WILL BE PERFORMED. THE CONTRACTOR SHALL CHECK ALL EXISTING CONDITIONS WHICH MAY AFFECT HIS WORK. NO ALLOWANCE WILL BE MADE FOR ANY EXTRA EXPENSE WHICH MAY BE INCURRED DUE TO NEGLIGENCE OR FAILURE TO EXAMINE THE SITE. EXISTING CONDITIONS, UTILITIES, ETC. THE SITE VISIT MUST BE DURING THE BID WEEK AND/OR PREPARED WITH THE OWNER. | CAL             | CALIFORNIA BUILDING CODE                       |
| 4.                                                                                                        | CONTRACTOR SHALL PROVIDE QUALIFIED TRADE PROFESSIONALS WITH ADEQUATE SAFETY TRAINING IN CONSTRUCTION OF THESE TYPES OF ACTIVITIES. SAFETY PRACTICES SHALL REQUIRE CONTRACTORS AT THEIR OWN DISCRETION.                                                                                                                                                                                                                                                                     | CEC             | CLEARING OUTLINE                               |
| 5.                                                                                                        | CONTRACTOR SHALL SCHEDULE AND COORDINATE THE FOLLOWING WITH OWNER OR OWNER'S PROJECT MANAGER.<br>A. ANY AND ALL UTILITIES SHOWN ON DRAWING, CONTRACTOR SHALL GIVE A MINIMUM OF TWO WEEKS PRIOR NOTICE FOR EACH SHUT DOWN.<br>B. ANY AND ALL WORK BEING PERFORMED IN AREAS WHERE THE FACILITY WILL REMAIN IN OPERATION.<br>C. ANY WORK WITHIN THE PARAMETER OF ITEMS A & B ABOVE SHALL BE COORDINATED WITH THE GENERAL CONTRACTOR.                                          | CFR             | CLEARING OUTLINE                               |
| 6.                                                                                                        | ALL MATERIALS AND WORKMANSHIP ARE SUBJECT TO APPROVAL BY OWNER. IF DEFECTIVE PORTION OF THE WORK SHALL BE REPLACED BY THE CONTRACTOR AT HIS PART OF THIS CONTRACT AT NO ADDITIONAL COST TO THE OWNER.                                                                                                                                                                                                                                                                      | CMU             | COMMON MASONRY UNIT                            |
| 7.                                                                                                        | Absolute accuracy of drawings cannot be guaranteed. While every effort has been made to coordinate the location of the existing ductwork and piping etc. it is the responsibility of the contractor to coordinate the exact requirements governed by actual job conditions at the time of construction.                                                                                                                                                                    | CNDWR           | CONDENSER WATER RETURN                         |
| 8.                                                                                                        | IN THE AREA OF NEW CONSTRUCTION WILL BE DONE A NUMBER OF EXISTING SERVICES. CONTRACTOR SHALL PROTECT AND MAINTAIN ALL ACTIVE LINES IN GOOD OPERATIONAL CONDITION.                                                                                                                                                                                                                                                                                                          | CNDWS           | CONDENSER WATER SUPPLY                         |
| 9.                                                                                                        | PROTECT EXISTING BUILDING STRUCTURES AND ADJACENT FINISHED SURFACES FROM DAMAGE BY WORK UNDER THIS DIVISION TO MATCH ADJACENT WORK. PROTECTING AND REPAIRING SHALL BE PERFORMED BY WORKMEN SKILLED IN THE TRADES INVOLVED.                                                                                                                                                                                                                                                 | CNG             | CONDENSER GAS                                  |
| 10.                                                                                                       | BALANCING CONTRACTOR SHALL BALANCE ALL EQUIPMENT SHOWN ON THESE DRAWINGS FOR FLOW QUANTITIES AND PRESSURE OR VACUUM INDICATED. BALANCING CONTRACTOR SHALL BE NEIGH CERTIFIED.                                                                                                                                                                                                                                                                                              | CNO             | CONDENSER OIL                                  |
| 11.                                                                                                       | ALL MECHANICALS INVOLVED IN PIPE AND DUCT DEMOLITION SHALL WEAR PROPER PERSONAL PROTECTIVE GEAR AND SHALL MAKE THE NECESSARY PROVISIONS TO PREVENT EXPOSURE TO HAZARDOUS MATERIALS.                                                                                                                                                                                                                                                                                        | COCG            | CONDENSER COOLING COIL                         |
| 12.                                                                                                       | RELIEF OR OPEN FLAME ARE NOT PERMITTED DURING THE DEMOLITION PROCESS OF ANY FLAMMABLE OR COMBUSTIBLE PIPING OR EXHAUST DUCT SYSTEMS.                                                                                                                                                                                                                                                                                                                                       | COE             | CONDENSER EXHAUST                              |
| 13.                                                                                                       | A JAIL WELDING PERMIT / HOT WORK SHALL BE OBTAINED FROM OWNER PRIOR TO COMMENCING ANY WELDING ACTIVITY.                                                                                                                                                                                                                                                                                                                                                                    | COF             | CONDENSER OIL FILL                             |
| 14.                                                                                                       | CONTRACTOR SHALL SUBMIT DUCT MATERIAL SUBMITTALS TO OWNER AND TO THE DESIGN ENGINEER FOR REVIEW AND APPROVAL, PRIOR TO PURCHASING AND INSTALLATION.                                                                                                                                                                                                                                                                                                                        | COH             | CONDENSER HEAT EXCHANGER                       |
| 15.                                                                                                       | CONTRACTOR SHALL COORDINATE FINAL INSPECTION OF COMPLETED WORK WITH DESIGN ENGINEER AND OWNER OR OWNER'S REPRESENTATIVE, FOR FINAL APPROVAL AND ACCEPTANCE.                                                                                                                                                                                                                                                                                                                | COI             | CONDENSER INLET                                |
| 16.                                                                                                       | ANY NEW OR EXISTING DUCTWORK OFFSETS REQUIRED AS A RESULT OF EXISTING JOB CONDITIONS OR LACK OF COORDINATION WITH OTHER TRADES, SHALL BE PROVIDED AT NO ADDITIONAL COST TO OWNER AND SUBJECT TO OWNER'S REVIEW.                                                                                                                                                                                                                                                            | COJ             | CONDENSER JOINT                                |
| 17.                                                                                                       | COORDINATE LOCATION OF EXISTING DUCTWORK AND NEW DUCTWORK OR PIPE RACKS WITH OTHER TRADES.                                                                                                                                                                                                                                                                                                                                                                                 | COL             | CONDENSER OIL LINE                             |
| 18.                                                                                                       | CONTRACTOR SHALL PROVIDE DUCTWORK AND TRANSITIONS EQUAL TO FREE AREA OF DUCTWORK AS SHOWN ON DRAWING. PROVIDE ANY ADDITIONAL DUCTWORK AS REQUIRED TO ADD NO ADDITIONAL COST TO OWNER AND SUBJECT TO OWNER'S REVIEW.                                                                                                                                                                                                                                                        | COM             | CONDENSER OIL MOTOR                            |
| 19.                                                                                                       | ALL DUCTWORK SHALL BE SUPPORTED ACCORDING TO SMACHA AND DEMONSTRATED BRACE PER SMACHA'S DESIGN RESTRAINT MANUAL TO MEET BUILDING CODE.                                                                                                                                                                                                                                                                                                                                     | CON             | CONDENSER OIL NUT                              |
| 20.                                                                                                       | MAINTAIN A MINIMUM HEAD CLEARANCE OF 84" ABOVE FINISHED FLOOR UNDER SUSPENDED DUCTWORK.                                                                                                                                                                                                                                                                                                                                                                                    | CONV            | CONDENSER OIL VALVE                            |
| 21.                                                                                                       | VERIFY WITH ENVIRONMENTAL HEALTH AND SAFETY DEPARTMENT (EH&S) THAT EXISTING DUCTWORK CONVEYING HAZARDOUS PRODUCTION MATERIALS (HPMS) HAS BEEN DECONTAMINATED AND CLEARED AS SUCH PRIOR TO CUTTING INTO OR DEMOLISHING. EH&S REPRESENTATIVE SHALL BE PRESENT WHEN CONTRACTOR TAKES INTO SUCH DUCTWORK OR FIRING.                                                                                                                                                            | COO             | CONDENSER OIL OIL                              |
| <b>SAFETY NOTES</b>                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | COQ             | CONDENSER OIL QUANTITY                         |
| 1.                                                                                                        | OPERATION OF ENERGY ISOLATING DEVICES SHALL BE PERFORMED BY SLAC WORKERS. CONTACT THE FIELD CONSTRUCTION MANAGER FOR ASSISTANCE WITH EQUIPMENT OPERATION.                                                                                                                                                                                                                                                                                                                  | COX             | CONDENSER OIL EXCHANGE                         |
| 2.                                                                                                        | SYSTEMS TOTALLY CONTAINING HAZARDOUS ENERGY SHALL BE LOCKED OUT IN ACCORDANCE WITH ESH SMACHA CHAPTER 51.                                                                                                                                                                                                                                                                                                                                                                  | COY             | CONDENSER OIL YIELD                            |
| <b>MECHANICAL SCOPE SUMMARY</b>                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | COZ             | CONDENSER OIL ZONE                             |
| 1.                                                                                                        | DEM0 B116 & B132 EXISTING LIBERTY FAN COIL AND OUTDOOR CONDENSING UNITS                                                                                                                                                                                                                                                                                                                                                                                                    | CPD             | CONDENSER OIL PUMP                             |
| 2.                                                                                                        | PARTIAL DEM0 B116 SUPPLY AIR DUCT AND FULL DEM0 OF RETURN SUPPLY AND RETURN GRILLE                                                                                                                                                                                                                                                                                                                                                                                         | CPF             | CONDENSER OIL PUMP FEED                        |
| 3.                                                                                                        | INSTALL NEW VERTIV FAN COIL IN B116 & B132 ELECTRICAL ROOM.                                                                                                                                                                                                                                                                                                                                                                                                                | CPH             | CONDENSER OIL PUMP HEAD                        |
| 4.                                                                                                        | INSTALL NEW DUCTWORK, CEILING SUPPLY GRILLES, AND RETURN GRILLE IN B116 & B132                                                                                                                                                                                                                                                                                                                                                                                             | CPJ             | CONDENSER OIL PUMP JOINT                       |
| <b>REFERENCE SYMBOLS</b>                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CPK             | CONDENSER OIL PUMP KICK                        |
| SUBJECT                                                                                                   | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CPQ             | CONDENSER OIL PUMP QUANTITY                    |
| DETAIL SHEET #                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CPR             | CONDENSER OIL PUMP RATE                        |
| EQUIPMENT TAG #                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CPS             | CONDENSER OIL PUMP SIZE                        |
| SCHEDULE                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CPV             | CONDENSER OIL PUMP VALVE                       |
| LOFT LIGHTING # WINDOW TAG #                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CPW             | CONDENSER OIL PUMP WEIGHT                      |
| FLOOR TAG #                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CPX             | CONDENSER OIL PUMP EXCHANGE                    |
| REVISION #                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CPY             | CONDENSER OIL PUMP YIELD                       |
| KEY NOTE #                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CPZ             | CONDENSER OIL PUMP ZONE                        |
| POINT OF CONNECTION TO EXISTING (POC)                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |                                                |
| ELEVATION DATA                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |                                                |
| EXISTING TO BE REMOVED (DEMOLISHED)                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |                                                |
| EXISTING TO REMAIN                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |                                                |
| NEW                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |                                                |
| SYMBOLS AND ABBREVIATIONS LISTED ARE FOR GENERAL USE. DISREGARD THOSE WHICH ARE NOT USED ON THE DRAWINGS. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |                                                |
| <b>DUCT INSULATION SCHEDULE TABLE PER CALIFORNIA TITLE 24 SECTION 120.4</b>                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                 |                                                |
| MANUFACTURER                                                                                              | TYPE                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MINIMUM R-VALUE |                                                |
| REFLECTOR OR ECODOL                                                                                       | BURLED DUCT WRAP                                                                                                                                                                                                                                                                                                                                                                                                                                                           | R-8             |                                                |

EXISTING MECHANICAL SCHEDULE - SHOWN FOR REFERENCE ONLY

| EXISTING CONDENSING UNIT / FAN COIL SCHEDULE |        |                 |                |                          |               |     |            |   |      |     |            |     |      |      |                 |                        |         |   |
|----------------------------------------------|--------|-----------------|----------------|--------------------------|---------------|-----|------------|---|------|-----|------------|-----|------|------|-----------------|------------------------|---------|---|
| TAG                                          | MAKE   | TYPE            | LOCATION       | LOCATION SERVED          | REFRIGERANT   |     | COMPRESSOR |   | COIL |     | ELECTRICAL |     |      |      | DIMENSIONS WxDH | OPERATING WEIGHT (LBS) | REMARKS |   |
|                                              |        |                 |                |                          | TYPE          | R   | QTY        | V | PH   | TH  | FLA        | MCA | MCCP |      |                 |                        |         |   |
| (D) B112-RCU-1001                            | LEBERT | FAN COIL        | BUEM6-CAG05547 | B112 METAL SHED BUILDING | B112 BPM ROOM | R22 | SINGLE     | 1 | 5    | 208 | 3          | 60  | 68   | 83.6 | 90              | 30-12"x30-12"x7H       | -       | - |
| (D) B112-RCU-1001                            | LEBERT | CONDENSING UNIT | C5083          | B112 AJO ADJACENT ROOF   | B119 BPM ROOM | R22 | N/A        | 1 | 5    | 208 | 3          | 60  | -    | -    | -               | -                      | -       | - |
| (D) B112-RCU-1001                            | LEBERT | FAN COIL        | BUEM6-CAG05547 | B112 METAL SHED BUILDING | B112 BPM ROOM | R22 | SINGLE     | 1 | 5    | 208 | 3          | 60  | 68   | 83.6 | 90              | 30-12"x30-12"x7H       | 870     | - |
| (D) B112-RCU-1001                            | LEBERT | CONDENSING UNIT | C5083          | B112 AJO ADJACENT ROOF   | B112 BPM ROOM | R22 | N/A        | 1 | 5    | 208 | 3          | 60  | -    | -    | -               | -                      | -       | - |

\* Connect refrigerant piping from condenser to respective fan coil. Field verify for exact routing, install per manufacturer's installation requirements, clearances, and refrigerant line limits (300 ft, consult factory if needed).  
\* Provide disconnect switch, #4 all equipment cords, LEDB PCBs network controller, and Modine cooled coil.  
\* Complete disconnect piping in respective outdoor condenser. Field verify for exact routing, install per manufacturer's installation requirements, clearances, and refrigerant line limits (300 ft, consult factory if needed).

## NEW MECHANICAL SCHEDULE

| NEW AIR DISTRIBUTION SCHEDULE |      |       |           |                                                       |         |
|-------------------------------|------|-------|-----------|-------------------------------------------------------|---------|
| SYMBOL                        | MFR  | MODEL | CFM       | DESCRIPTION                                           | REMARKS |
| CSD                           | TTUS | MCD   | 1000/1200 | 24X24 CEILING SUPPLY AIR DIFFUSER, 24X24 SQUARE NECK. | 1       |
| SWR-1                         | TTUS | SSFL  | 1600      | 30X18 SIDE WALL RETURN AIR GRILLE.                    | -       |
| SWR-2                         | TTUS | SSFL  | 3200      | 30X30 SIDE WALL RETURN AIR GRILLE.                    | -       |

REMARKS:  
 PROVIDE RAPID MOUNT FRAME AND OPPOSED BLADE DAMPER.

| NEW CONDENSING UNIT / FAN COIL SCHEDULE                                                                                                                                                                                                 |      |               |                 |       |                          |                 |                  |                          |                             |            |         |     |             |             |                 |                      |            |                   |                         |                         |                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|---------------|-----------------|-------|--------------------------|-----------------|------------------|--------------------------|-----------------------------|------------|---------|-----|-------------|-------------|-----------------|----------------------|------------|-------------------|-------------------------|-------------------------|-------------------|
| UNIT                                                                                                                                                                                                                                    | TAQ  | MAKE          | TYPE            | MODEL | LOCATION                 | LOCATION SERVED | REFRIGERANT TYPE | REFRIGERANT CHARGE (LBS) | REFRIGERANT CONNECTION SIZE | COMPRESSOR | Q1 TONS | QFM | ESP (IN WG) | TSP (IN WG) | CLG CAP TOT MBH | SENS CLG CAP TOT MBH | ELECTRICAL | DIMENSIONS INxHxI | OPERATING WEIGHT (LBS)  | REMARKS                 |                   |
| (N 8.152-TCU)                                                                                                                                                                                                                           | 1001 | VERTIV/LEBERT | FAN COIL        | PC609 | B116 METAL SHED BUILDING | B116 BPH ROOM   | R454B            | 0.9                      | 3/4" 1-1/8"                 | SINGLE     | 1       | 8   | 3200        | 0.28        | 0.75            | 155.3                | 91.4       | 208 3             | 60 75.7 100             | 34-10 3/4x34-3/8x77-1/2 | 700 3.4, 5.7, 8.7 |
| (N 8.152-RCU)                                                                                                                                                                                                                           | 1001 | VERTIV/LEBERT | CONDENSING UNIT | PC609 | B116 PARKING LOT         | B116 BPH ROOM   | R454B            | 1                        | 3/4" 1-1/8"                 | SINGLE     | 1       | 8   | -           | -           | -               | -                    | 208 3      | 60 75.7 100       | 34-10 3/4x34-3/8x77-1/2 | 700 3.4, 5.7, 8.7       |                   |
| (N 8.152-TCU)                                                                                                                                                                                                                           | 1001 | VERTIV/LEBERT | FAN COIL        | PC609 | B132 METAL SHED BUILDING | B132 BPH ROOM   | R454B            | 0.9                      | 3/4" 1-1/8"                 | SINGLE     | 1       | 8   | 3200        | 0.25        | 0.75            | 155.9                | 91.4       | 208 3             | 60 75.7 100             | 34-10 3/4x34-3/8x77-1/2 | 700 3.4, 5.7, 8.7 |
| (N 8.152-RCU)                                                                                                                                                                                                                           | 1001 | VERTIV/LEBERT | CONDENSING UNIT | PC609 | B132 PARKING LOT         | B132 BPH ROOM   | R454B            | 1                        | 3/4" 1-1/8"                 | SINGLE     | 1       | 8   | -           | -           | -               | -                    | 208 3      | 60 75.7 100       | 34-10 3/4x34-3/8x77-1/2 | 700 3.4, 5.7, 8.7       |                   |
| Remarks:                                                                                                                                                                                                                                |      |               |                 |       |                          |                 |                  |                          |                             |            |         |     |             |             |                 |                      |            |                   |                         |                         |                   |
| 1. Connect refrigerant piping from condenser to respective fan coil. Field verify for exact routing. Install per manufacturer's installation requirements, clearances, and refrigerant line limits (300 ft. consult factory if needed). |      |               |                 |       |                          |                 |                  |                          |                             |            |         |     |             |             |                 |                      |            |                   |                         |                         |                   |
| 2. Provide disconnect switch, 10/2700V <sup>0</sup> F equipment switch, Label POU network connection, and Modine isolated coil.                                                                                                         |      |               |                 |       |                          |                 |                  |                          |                             |            |         |     |             |             |                 |                      |            |                   |                         |                         |                   |
| 3. Connect refrigerant piping to respective outdoor condenser. Field verify for exact routing. Install per manufacturer's installation requirements, clearances, and refrigerant line limits (300 ft. consult factory if needed).       |      |               |                 |       |                          |                 |                  |                          |                             |            |         |     |             |             |                 |                      |            |                   |                         |                         |                   |
| 4. Provide 2-stage Electric Run Switch.                                                                                                                                                                                                 |      |               |                 |       |                          |                 |                  |                          |                             |            |         |     |             |             |                 |                      |            |                   |                         |                         |                   |
| 5. Refrigerant piping by manufacturer.                                                                                                                                                                                                  |      |               |                 |       |                          |                 |                  |                          |                             |            |         |     |             |             |                 |                      |            |                   |                         |                         |                   |
| 6. Provide disposable 34x24x1 MERV 11 filter.                                                                                                                                                                                           |      |               |                 |       |                          |                 |                  |                          |                             |            |         |     |             |             |                 |                      |            |                   |                         |                         |                   |
| 7. Provide factory dual smoke detector for local automatic shut-down.                                                                                                                                                                   |      |               |                 |       |                          |                 |                  |                          |                             |            |         |     |             |             |                 |                      |            |                   |                         |                         |                   |

## REFRIGERANT PIPING DIAGRAM (FOR REFERENCE ONLY)

[illegible]

| REFRIGERANT CALCULATIONS |                    |                          |             |                       |               |                                                      |                                          |                                   |           |
|--------------------------|--------------------|--------------------------|-------------|-----------------------|---------------|------------------------------------------------------|------------------------------------------|-----------------------------------|-----------|
| CONDENSING UNIT          | FAN COIL           | ROOM                     | AREA (FT²)* | ESTIMATED HEIGHT (FT) | VOLUME (FT³)* | ASHRAE 15 R454B ALLOWABLE REFRIGERANT CHARGE LB/FT³* | ESTIMATED SYSTEM REFRIGERANT CHARGE (LB) | ALLOWABLE REFRIGERANT CHARGE (LB) | COMPLETS? |
| (N) B-116-RCU-1001       | (N) B-116-FCU-1001 | B116 META. SHED BUILDING | 597         | 18.0                  | 10748.0       | 3.1                                                  | 6.9                                      | 33.3                              | YES       |
| (N) B-132-RCU-1001       | (N) B-132-FCU-1001 | B132 META. SHED BUILDING | 525         | 18.0                  | 9450.0        | 3.1                                                  | 6.9                                      | 29.3                              | YES       |

SAFETY NOTES

1. OPERATION OF ENERGY ISOLATING DEVICES SHALL BE PERFORMED BY SLAC WORKERS. CONTACT THE FIELD CONSTRUCTION MANAGER FOR ASSISTANCE WITH EQUIPMENT OPERATION.

2. SYSTEMS TOTALLY CONTAINING HAZARDOUS ENERGY SHALL BE LOCKED OUT IN ACCORDANCE WITH SLAC ESH MANUAL CHAPTER 51.

MECHANICAL SCOPE SUMMARY

DEMO B116 & B132 EXISTING LIEBERT FAN COIL AND OUTDOOR CONDENSER UNITS.

PARTIAL DEMO B116 SUPPLY AIR DUCT AND FULL DEMO OF B116 SUPPLY AND RETURN AIR DUCTS.

INSTALL NEW VERTIV FAN COIL IN B116 & B132 ELECTRICAL ROOM.

INSTALL NEW DUCTWORK, CEILING SUPPLY GRILLES, AND RETURN GRILLES IN B116 & B132.

REFERENCE SYMBOLS

| SYMBOL | DESCRIPTION                           |
|--------|---------------------------------------|
|        | DETAIL #<br>SHEET #                   |
|        | SECTION #<br>SCHEDULE                 |
|        | EQUIPMENT TAG #                       |
|        | WINDOW TAG #                          |
|        | RECESS TAG #                          |
|        | REVISION #                            |
|        | KEY NOTE #                            |
|        | POINT OF CONNECTION TO EXISTING (POC) |
|        | ELEVATION DATUM                       |
|        | EXISTING TO BE REMOVED (DEMOLISHED)   |
|        | EXISTING TO REMAIN                    |
|        | NEW                                   |

SYMBOLS AND ABBREVIATIONS LISTED ARE FOR GENERAL USE. DISREGARD THOSE WHICH ARE NOT USED ON THE DRAWINGS.

DUCT INSULATION SCHEDULE TABLE PER CALIFORNIA TITLE 24 SECTION 120.4

| MANUFACTURER        | TYPE             | MINIMUM R-VALUE |
|---------------------|------------------|-----------------|
| REFLECTIV OR SECOND | BUBBLE DUCT WRAP | R-8             |

| DUCT SYMBOLS |             |                                           |
|--------------|-------------|-------------------------------------------|
| SINGLE LINE  | DOUBLE LINE | DESCRIPTION                               |
|              |             | SUPPLY AIR DUCT UP OR SECTION             |
|              |             | RETURN AIR DUCT UP OR SECTION             |
|              |             | EXHAUST AIR DUCT UP OR SECTION            |
|              |             | SUPPLY AIR DUCT DOWN                      |
|              |             | RETURN AIR DUCT DOWN                      |
|              |             | EXHAUST AIR DUCT DOWN                     |
|              |             | CEILING SUPPLY DIFFUSER OR REGISTER       |
|              |             | CEILING RETURN AIR GRILLE                 |
|              |             | CEILING EXHAUST AIR GRILLE                |
|              |             | EXL WALL EXHAUST AIR GRILLE               |
|              |             | RA WALL RETURN AIR GRILLE                 |
|              |             | SA WALL SUPPLY AIR DIFFUSER OR GRILLE     |
|              |             | FAN POWER HEPA FILTER                     |
|              |             | GRILL VENTILATED KNOCKED LEAF HEPA FILTER |
|              |             | W/FACE VELOCITY SUPPLYING SEC. UNIT       |
|              |             | GRILL VENTILATED KNOCKED LEAF HEPA FILTER |
|              |             | W/FACE VELOCITY SUPPLYING SEC. UNIT       |
|              |             | DUCT WITH HANGING                         |
|              |             | DUCT RISE OR DROP IN DIRECTION OF AIRFLOW |
|              |             | DUCT RISE OR DROP IN DIRECTION OF AIRFLOW |
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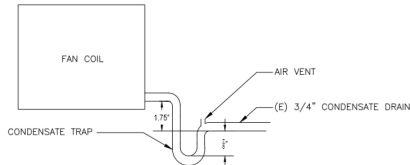
| SLAC BUILDING INSPECTION OFFICE                                                                                                                       |              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Construction Authorization (FC)                                                                                                                       |              |
| BY: onslar                                                                                                                                            |              |
| DATE: 1/23/2020                                                                                                                                       | PRJ#: 25-254 |
| Revised Authorization <input checked="" type="checkbox"/> No <input type="checkbox"/> Yes                                                             |              |
| These plans have been reviewed for compliance with applicable building codes and standards, and EIR program requirements.                             |              |
| The authorization of these plans shall not be construed to be a permit or approval for violation of any applicable codes, standards, or EIR programs. |              |
| <b>THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.</b>                                                                    |              |

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

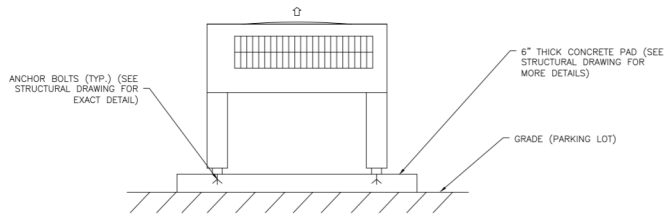


|                                                                                                                                                                             |                                         |                      |                                                                       |                         |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|----------------------|-----------------------------------------------------------------------|-------------------------|--------|
| SCALE AS NOTED                                                                                                                                                              |                                         | DO NOT SCALE DRAWING | 55010.2025.01                                                         |                         | M.0.01 |
|   |                                         |                      | B116 AND B132<br>B116 & B132 HVAC DESIGN<br>MECHANICAL<br>COVER SHEET |                         |        |
| THE DRAWINGS, SPECIFICATIONS AND DATA HEREIN PROVIDED SHALL NOT BE USED FOR ANY PURPOSES OTHER THAN SUBMITTED WITHOUT THE WRITTEN PERMISSION OF STAFF AND UNIVERSITY, SLAC  |                                         |                      |                                                                       |                         |        |
| EFOR: 0000-00000000<br>DESIGNED BY: J. J. JAMES<br>CHECKED BY: J. J. JAMES                                                                                                  | DATE: 01/15/2025<br>EFOR: 0000-00000000 | DATE: 01/15/2025     | DRAWING NUMBER:<br>B0116-M00-00002                                    | REVISION NUMBER:<br>D E |        |

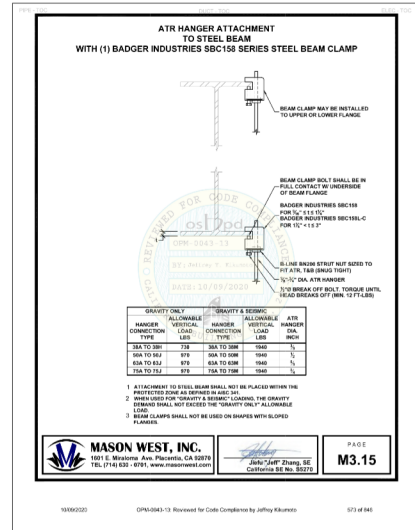
| REV | DESCRIPTION | DRN | CHK | APP | DATE       |
|-----|-------------|-----|-----|-----|------------|
| A   | 50% CD      | KT  | JL  | MHC | 10-02-2025 |
| B   | 100% CD     | KT  | JL  | MHC | 10-27-2025 |
| C   | FAO REVIEW  | KT  | JL  | MHC | 11-21-2025 |
| D   | BIO REVIEW  | KT  | JL  | MHC | 12-16-2025 |



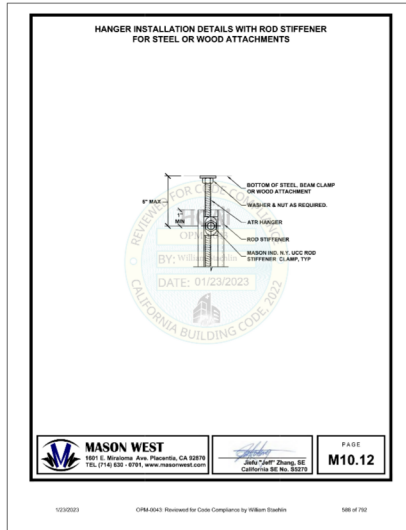
① CONDENSATE DRAIN TRAP DETAIL  
SCALE: N.T.S.



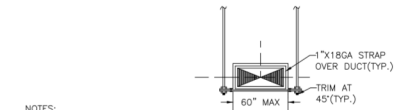
② FLOOR MOUNTED CONDENSER DETAIL  
SCALE: N.T.S.



③ STEEL BEAM CLAMP DETAIL  
SCALE: N.T.S.



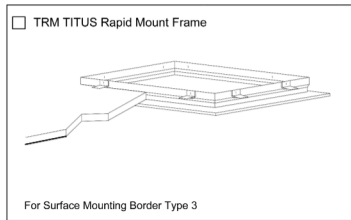
④ ROD STIFFENER ATTACHMENT DETAIL  
SCALE: N.T.S.



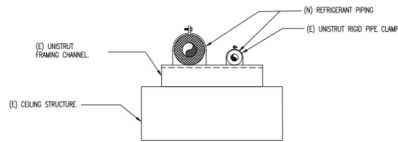
- NOTES:
- HANGER SPACING WILL BE 6'.
  - SWAY & SEISMIC BRACING IS NOT REQUIRED PER ASCE 7-16 EXCEPTION LISTED BELOW.  
2.1.1. THE DUCT HAS A CROSS-SECTIONAL AREA LESS THAN 6.0 SQUARE FEET.  
2.1.2. THE DUCT WEIGHS 20 POUNDS PER FOOT OR LESS.
  - HANGER MATERIAL SUPPORTING FLEXIBLE DUCT SHALL IN NO CASE BE LESS THAN 1/2" WIDE. FLEXIBLE DUCT SHALL BE SUPPORTED PER MANUFACTURER RECOMMENDED MATERIALS, BUT AT NO GREATER DISTANCE THAN 4FT MAX. PERMISSIBLE SAG IS MAX. 1" PER FOOT OF SPACING BETWEEN SUPPORTS.

| RECTANGULAR DUCT HANGERS MINIMUM SIZE |                        |                            |                           |                        |
|---------------------------------------|------------------------|----------------------------|---------------------------|------------------------|
| MAX DUCT<br>LOAD (POUNDS)             | DUCT<br>SPACING (FEET) | MIN DUCT<br>SPACING (FEET) | MAX DUCT<br>LOAD (POUNDS) | DUCT<br>SPACING (FEET) |
| 1/2" x 1/2"                           | 1' x 1/2"              | 1' x 1/2"                  | 1' x 1/2"                 | 1' x 1/2"              |
| 1/2" x 3/4"                           | 1' x 3/4"              | 1' x 3/4"                  | 1' x 3/4"                 | 1' x 3/4"              |
| 1/2" x 1"                             | 1' x 1"                | 1' x 1"                    | 1' x 1"                   | 1' x 1"                |
| 1/2" x 1 1/4"                         | 1' x 1 1/4"            | 1' x 1 1/4"                | 1' x 1 1/4"               | 1' x 1 1/4"            |

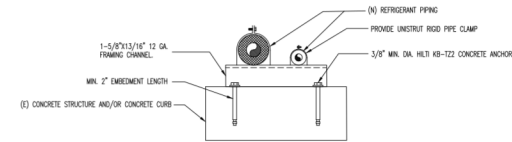
⑤ TYPICAL DUCT HANGER DETAIL  
SCALE: N.T.S.



⑥ EXISTING HARD CEILING DIFFUSER MOUNTING FRAME DETAIL  
SCALE: N.T.S.



⑦ EXISTING REFRIGERANT PIPE SUPPORT DETAIL  
SCALE: N.T.S.



⑧ NEW EXTERIOR REFRIGERANT PIPE SUPPORT DETAIL  
SCALE: N.T.S.

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

|                                                                                                                                                                                         |             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| SLAC BUILDING INSPECTION OFFICE                                                                                                                                                         |             |
| Construction Authorization (PC)                                                                                                                                                         |             |
| BY: <input type="checkbox"/> order                                                                                                                                                      |             |
| DATE: 10/20/2025                                                                                                                                                                        | PRJ: 25-204 |
| Revised Authorization: <input checked="" type="checkbox"/> No <input type="checkbox"/> Yes                                                                                              |             |
| These plans have been reviewed and approved for construction by the SLAC Building Inspection Office. The information on these plans shall not be used for any other project or purpose. |             |
| THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.                                                                                                             |             |



|                                |  |                      |  |                        |  |                                                            |  |
|--------------------------------|--|----------------------|--|------------------------|--|------------------------------------------------------------|--|
| SCALE: AS NOTED                |  | DO NOT SCALE DRAWING |  | S5010 2025.01          |  | M0.02                                                      |  |
| ENGR: CHEN/UNISTRUT            |  | DATE: 10/20/2025     |  | APPROVALS: [Signature] |  | DATE: 10/20/2025                                           |  |
| DES: NAK/CEIN                  |  | DATE: 10/20/2025     |  | APPROVALS: [Signature] |  | DATE: 10/20/2025                                           |  |
| CHK: L/AMT                     |  | DATE: 10/20/2025     |  | APPROVALS: [Signature] |  | DATE: 10/20/2025                                           |  |
| DRAWING NUMBER: B0116-M05-0003 |  | REVISION: D          |  | E                      |  | B116 AND B132 B116 AND B132 HVAC Design MECHANICAL DETAILS |  |



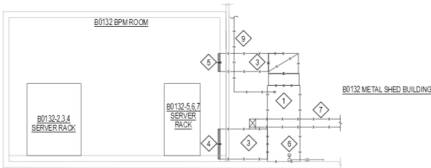
| REV | DESCRIPTION | DRN | CHK | APP | DATE       |
|-----|-------------|-----|-----|-----|------------|
| A   | 50% CD      | KT  | JL  | MHC | 10-02-2025 |
| B   | 100% CD     | KT  | JL  | MHC | 10-27-2025 |
| C   | FAO REVIEW  | KT  | JL  | MHC | 11-21-2025 |
| D   | BIO REVIEW  | KT  | JL  | MHC | 12-16-2025 |

PLAN GENERAL NOTES

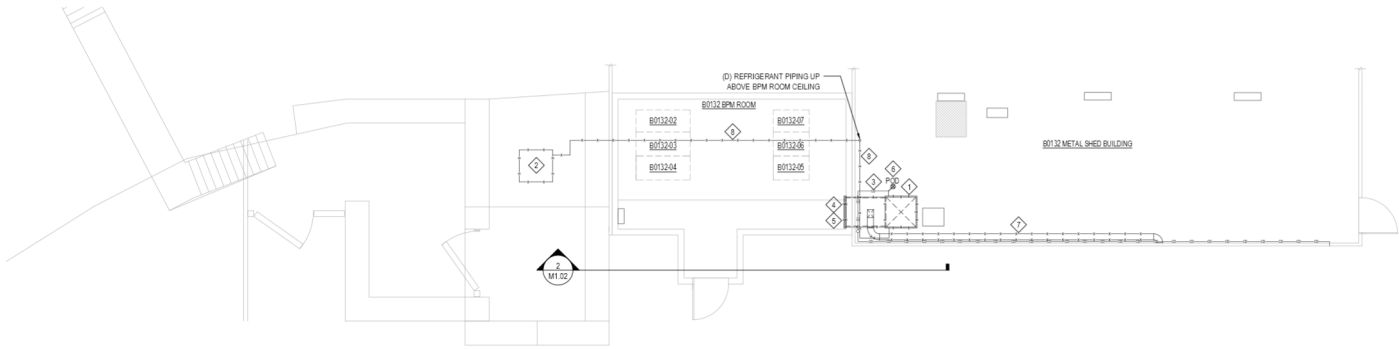
- EXISTING MECHANICAL TO REMAIN, UNLESS OTHERWISE NOTED.
- SEE ELECTRICAL DRAWINGS FOR ALL ELECTRICAL WORK.

MECHANICAL SHEET NOTES

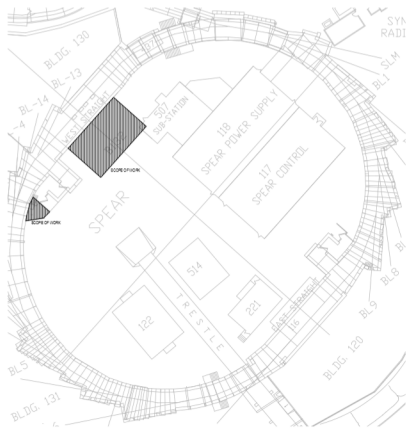
- DEMO EXISTING 5-TON INDOOR LIEBERT FANCOIL.
- DEMO EXISTING 5-TON OUTDOOR LIEBERT CONDENSING UNIT ON ROOF.
- DEMO EXISTING SUPPLY DUCTWORK ABOVE AND RETURN DUCTWORK BELOW.
- DEMO EXISTING FLOOR LEVEL 30X30 SIDE RETURN AIR GRILLE. PATCH AND FILL WALL WITH MATERIAL MATCHING EXISTING CONDITIONS.
- DEMO EXISTING CEILING LEVEL 30X18 SIDE WALL SUPPLY AIR GRILLE.
- PARTIAL DEMO EXISTING 3/4" CONDENSATE DRAIN FROM EXISTING FAN COIL AS SHOWN ON PLAN.
- DEMO EXISTING 8X8 OUTSIDE AIR DUCTWORK. PATCH AND FILL WALL PENETRATION TO EXTERIOR.
- DEMO EXISTING REFRIGERANT PIPING BETWEEN EXISTING FAN COIL AND OUTDOOR CONDENSING UNIT.



2 B0132 - MECHANICAL SECTION DEMO PLAN  
SCALE: 1/4" = 1'-0"



1 B0132 - MECHANICAL DEMO PLAN  
SCALE: 1/4" = 1'-0"



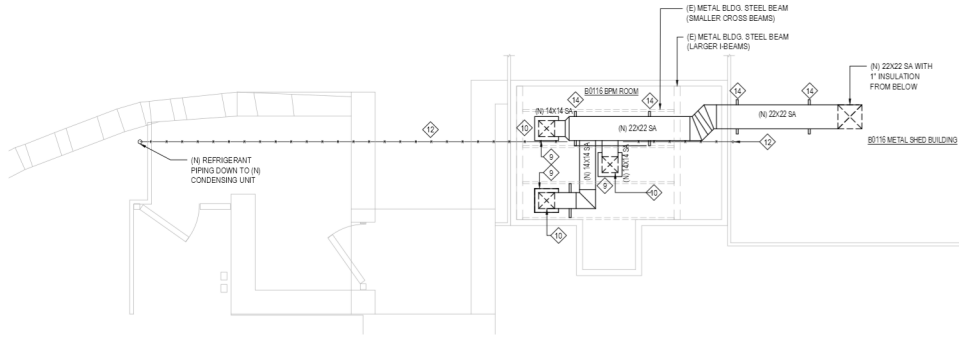
3 SITE PLAN  
SCALE: NTS

All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

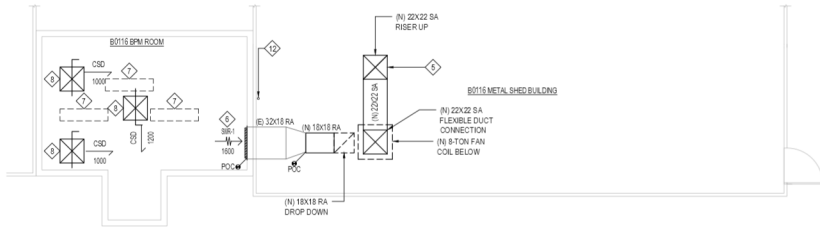
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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| SLAC BUILDING INSPECTION OFFICE                                                                                                                                                                 |                                                                   |
| Construction Authorization (PC)                                                                                                                                                                 |                                                                   |
| BY: <input type="checkbox"/> <input checked="" type="checkbox"/>                                                                                                                                | PRM: <input type="checkbox"/> <input checked="" type="checkbox"/> |
| DATE: 10/20/2025                                                                                                                                                                                | PRM: 25-204                                                       |
| Revised Authorization <input checked="" type="checkbox"/> No <input type="checkbox"/> Yes                                                                                                       |                                                                   |
| These plans have been reviewed and approved for construction by the SLAC Building Inspection Office. The authorization of these plans shall not be construed as a warranty of any kind by SLAC. |                                                                   |
| These plans shall be kept on the job site for all construction inspections.                                                                                                                     |                                                                   |



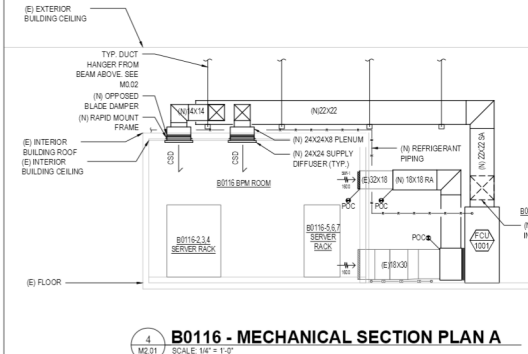
|                                                                                                                                                                        |                                                                   |                                                              |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------|---------------|
| SCALE: AS NOTED                                                                                                                                                        | DO NOT SCALE DRAWING                                              | 5/5010 2025.01                                               | M1.02         |
| SLAC ENERGY                                                                                                                                                            |                                                                   | B132<br>B116 AND B132 HVAC Design<br>MECHANICAL<br>DEMO PLAN |               |
| THE DRAWINGS, SPECIFICATIONS AND OTHER DATA HEREIN PROVIDED SHALL NOT BE COPIED, PUBLISHED OR OTHERWISE FURTHER DISSEMINATED WITHOUT PRIOR WRITTEN PERMISSION OF SLAC. |                                                                   | DRAWING NUMBER<br>B0116-MD1-0005                             |               |
| ENGR: CHEN, L. CHEN<br>DES: N. K. CHEN<br>CHK: L. CHEN                                                                                                                 | DATE: 10/20/2025<br>APPROVALS: 10/20/2025<br>REVISION: 10/20/2025 | DATE: 10/20/2025                                             | REVISION: D E |



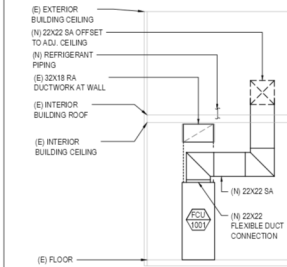
**3 B0116 - MECHANICAL ABOVE CEILING PLAN**  
SCALE: 1/4" = 1'-0"



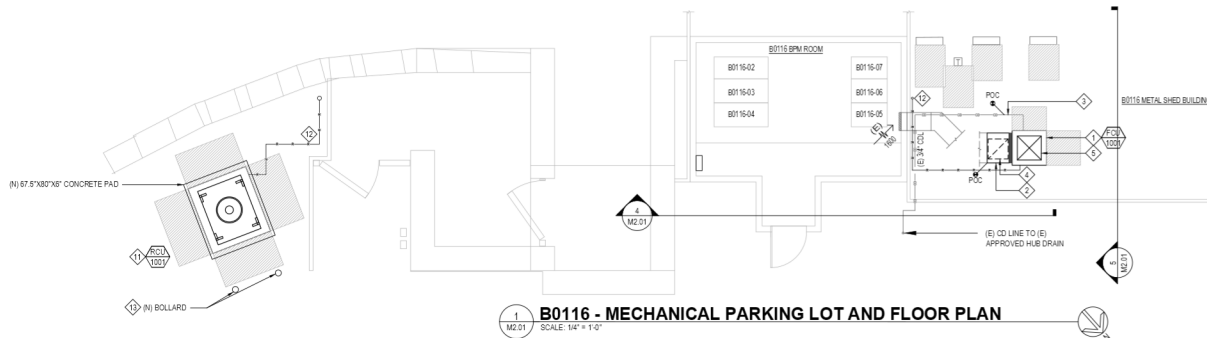
**2 B0116 - MECHANICAL RCP PLAN**  
SCALE: 1/4" = 1'-0"



**4 B0116 - MECHANICAL SECTION PLAN A**  
SCALE: 1/4" = 1'-0"



**5 B0116 - MECHANICAL SECTION PLAN B**  
SCALE: 1/4" = 1'-0"



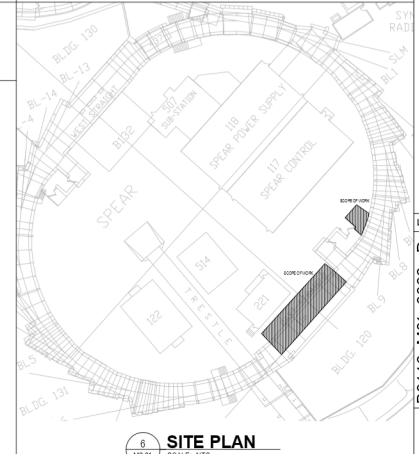
**1 B0116 - MECHANICAL PARKING LOT AND FLOOR PLAN**  
SCALE: 1/4" = 1'-0"

#### PLAN GENERAL NOTES

- EXISTING MECHANICAL TO REMAIN, UNLESS OTHERWISE NOTED.
- SEE ELECTRICAL DRAWINGS FOR ALL ELECTRICAL WORK.
- PROVIDE MANUAL VOLUME BALANCING DAMPERS AT EACH BRANCH TAKE-OFF IN ALL RETURN DUCTWORK. LOCATE BALANCING DAMPERS AS CLOSE TO THE BRANCH TAKE-OFF AS POSSIBLE.
- CONTRACTOR SHALL PROVIDE DUCT MATERIAL SUBMITTALS TO OWNER AND TO THE DESIGN ENGINEER FOR REVIEW AND APPROVAL, PRIOR TO PURCHASING AND INSTALLATION.
- ANY NEW OR EXISTING DUCTWORK OFFSETS REQUIRED AS A RESULT OF EXISTING JOB CONDITIONS OR LACK OF COORDINATION WITH OTHER TRADES, SHALL BE PROVIDED AT NO ADDITIONAL COST TO OWNER AND SUBJECT TO OWNER'S REVIEW.
- CONTRACTOR SHALL PROVIDE DUCTWORK AND TRANSITIONS EQUAL TO FREE AREA OF DUCTWORK AS SHOWN ON DRAWING, TO PREVENT A CONFLICT WITH EXISTING CONDITIONS.
- ALL DUCTWORK SHALL BE SUPPORTED ACCORDING TO SMACNA AND SEISMICALLY BRACED PER SMACNA'S SEISMIC RESTRAINT MANUAL TO MEET BUILDING CODE.
- TAB CONTRACTOR SHALL BALANCE ALL EQUIPMENT SHOWN ON THIS DOCUMENT PER THE FLOW QUANTITIES.
- CONTRACTOR SHALL COORDINATE WITH FINAL INSPECTION OR COMPLETED WORK WITH DESIGN ENGINEER AND OWNER OR OWNER'S REPRESENTATIVE, FOR FINAL APPROVAL AND ACCEPTANCE.
- CONTRACTOR SHALL VERIFY ALL EXISTING BUILDING CONDITIONS.

#### MECHANICAL SHEET NOTES

- PROVIDE NEW LIEBERT P1020U 8-TON UPFLOW UNIT. INSTALL 120" (+20" FURTHER) FROM THE DISCHARGE WALL THAN THE OLD FAN COIL LOCATION. MAINTAIN CODE REQUIRED CLEARANCES TO NEARBY ELECTRICAL PANEL AND EQUIPMENT. PROVIDE DUCT SMOKE DETECTOR FOR LOCAL AUTOMATIC SHUT-DOWN. FAN COIL SHALL BE LOCALLY CONTROLLED AND MAINTAIN 10" (+15" FIELD VERIFY FOR ANY OBSTRUCTION. PROVIDE FLOOR MOUNTING PER STRUCTURAL DRAWINGS.
- PROVIDE NEW 30X32X22 RETURN AIR PLENUM WITH FLEXIBLE CONNECTION TO RETURN AIR INLET OF FANCOIL UNIT. COMPLETE WITH DUCT INSULATION.
- PROVIDE NEW 3/4" CONDENSATE DRAIN LINE FROM FANCOIL, AND EXTEND AND RE-CONNECT TO EXISTING 3/4" CONDENSATE DRAIN. TERMINATE TO (E) HUB DRAIN AS SHOWN ON PLAN. FIELD VERIFY FOR EXACT LOCATIONS.
- EXISTING 32X18 RETURN AIR DUCT AT CEILING. PROVIDE NEW 18X18 RETURN AIR DUCT AND RE-CONNECT TO EXISTING. OFFSET DUCT AND DROP DOWN INTO NEW 30X32X22 RETURN AIR PLENUM. PROVIDE DUCT INSULATION TO EXISTING AND NEW RETURN AIR DUCTWORK.
- PROVIDE NEW 22X22 SUPPLY AIR DUCT FROM FANCOIL OUTLET AND OFFSET TO ADJACENT ROOM CEILING. PROVIDE FLEXIBLE DUCT CONNECTION AT FANCOIL OUTLET. PROVIDE DUCT INSULATION AND UNISTRUT DUCT SUPPORTS FROM CEILING STEEL BEAMS. SEE DETAILS ON M02.
- PROVIDE NEW 30X18 SIDEWALL RETURN AIR GRILLE AT EXISTING WALL OPENING CLOSE TO CEILING.
- EXISTING CEILING LIGHTS TO REMAIN. COORDINATE WITH ELECTRICAL AND FIELD VERIFY FOR EXACT LOCATION.
- PROVIDE NEW 4-WAY ADJUSTABLE 24X24 CEILING SUPPLY AIR DIFFUSER. COMPLETE WITH OPPOSED BLADE DAMPER AND RAPID MOUNT CEILING FRAME.
- PROVIDE NEW 24X24X8 PLENUM ABOVE 24X24 CEILING SUPPLY AIR DIFFUSER AND CONNECT TO NEW SUPPLY AIR BRANCH DUCT.
- PROVIDE NEW 14X14 SUPPLY AIR BRANCH DUCT. PROVIDE DUCT INSULATION AND UNISTRUT DUCT SUPPORTS FROM CEILING STEEL BEAMS. SEE DETAILS ON M02.
- PROVIDE NEW LIEBERT MCL055 OUTDOOR CONDENSING UNIT ON 67.5'X80X6" CONCRETE PAD. INSTALL PER MANUFACTURER'S INSTALLATION REQUIREMENTS AND CLEARANCES. EQUIPMENT IS LOCATED IN A SECURE LOCATION. COORDINATE WITH ELECTRICAL FOR POWER CONNECTION. REFER TO STRUCTURAL DRAWINGS FOR MOUNTING DETAILS.
- NEW REFRIGERANT PIPING BETWEEN NEW FAN COIL AND OUTDOOR CONDENSING UNIT. RE-USE EXISTING UNISTRUT MOUNT. PROVIDE NEW UNISTRUT MOUNT ABOVE EXISTING CONCRETE ROOF AS REQUIRED. REFRIGERANT PIPING EXPOSED TO WEATHER SHALL BE PROVIDED WITH A WATER RESISTANT COVER OVER THE INSULATION. ADHESIVE TRAP SHALL NOT BE USED FOR PROTECTION. FIELD VERIFY FOR EXACT EXISTING UNISTRUT LOCATIONS, ROUTING, AND CONDITIONS.
- NEW 6"Ø BOLLARD. COORDINATE WITH SLAC TO CONFIRM EXACT NUMBER OF BOLLARDS AND LOCATION.
- NEW UNISTRUT DUCT SUPPORT EVERY 6' FROM STEEL BEAM ABOVE. PROVIDE BEAM CLAMP, 3/8"Ø THREADED ROD, ROD STIFFENER, AND UNISTRUT SUPPORTS. SEE DETAILS ON M02. FIELD VERIFY EXISTING STEEL BEAM LOCATIONS PRIOR TO INSTALLATION.



**6 SITE PLAN**  
SCALE: NTS

All changes from the B10 approved job set of plans and specifications shall be submitted to and approved by the B10 office prior to the changes being made in the field.

|                                                                                                                                                                                                                               |             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| SLAC BUILDING INSPECTION OFFICE                                                                                                                                                                                               |             |
| Construction Authorization (B10)                                                                                                                                                                                              |             |
| BY: <input type="checkbox"/> order                                                                                                                                                                                            |             |
| DATE: 10/2/2020                                                                                                                                                                                                               | PRN: 25-204 |
| Revised Authorization: <input type="checkbox"/> No <input type="checkbox"/> Yes                                                                                                                                               |             |
| These plans have been reviewed and approved for construction by the SLAC Building Inspection Office. The authorization of these plans does not constitute a warranty or endorsement of any products, materials, or equipment. |             |
| THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.                                                                                                                                                   |             |

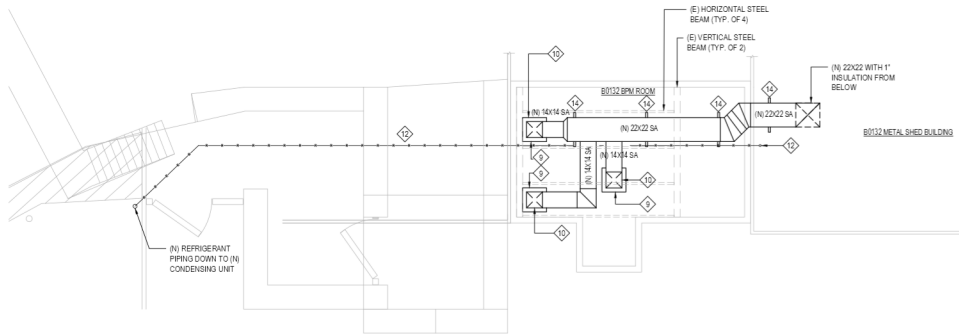


SCALE: AS NOTED DO NOT SCALE DRAWING S5010 20/25 01

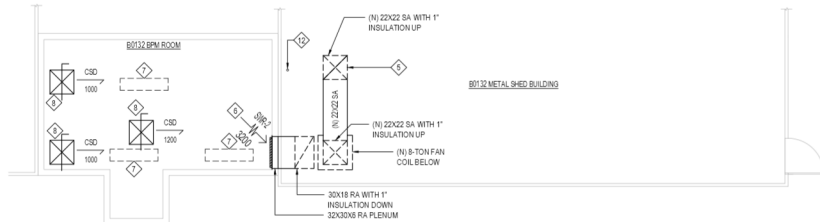
|                                                                                                                                                                                                        |                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| SLAC U.S. DEPARTMENT OF ENERGY                                                                                                                                                                         |                                               |
| THE DRAWINGS, SPECIFICATIONS AND OTHER DATA HEREIN PROVIDED SHALL NOT BE COPIED, PUBLISHED OR OTHERWISE FURTHER DISSEMINATED WITHOUT PRIOR WRITTEN PERMISSION OF SLAC NATIONAL ACCELERATOR LABORATORY. |                                               |
| ENGR: <input type="checkbox"/> 244/10/20/20                                                                                                                                                            | DATE: 10/2/2020                               |
| DES: <input type="checkbox"/> 244/10/20/20                                                                                                                                                             | APPROVALS: <input type="checkbox"/> 10/2/2020 |
| CHK: <input type="checkbox"/> 10/2/2020                                                                                                                                                                | DATE: 10/2/2020                               |

|                                                |               |
|------------------------------------------------|---------------|
| B116 B116 AND B132 HVAC Design MECHANICAL PLAN |               |
| DRAWING NUMBER: B0116-M01-0006                 | REVISION: D E |

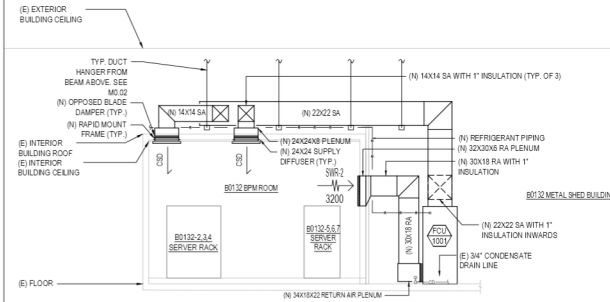
B0116-M01-0006 D E



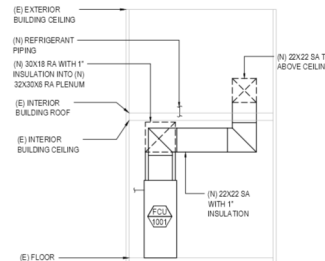
**B0132 - MECHANICAL ABOVE CEILING PLAN**  
SCALE: 1/4" = 1'-0"



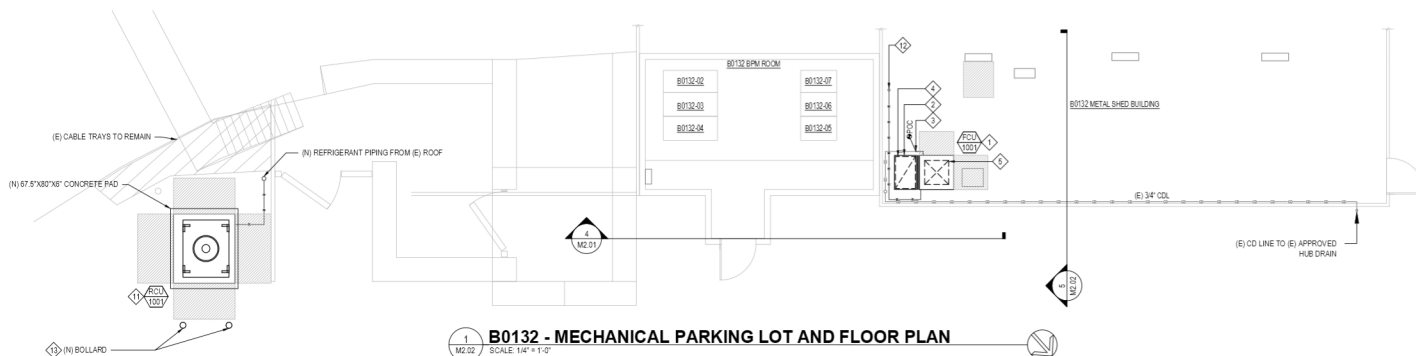
**B0132 - MECHANICAL RCP PLAN**  
SCALE: 1/4" = 1'-0"



**B0132 - MECHANICAL SECTION PLAN A**  
SCALE: 1/4" = 1'-0"



**B0132 - MECHANICAL SECTION PLAN B**  
SCALE: 1/4" = 1'-0"



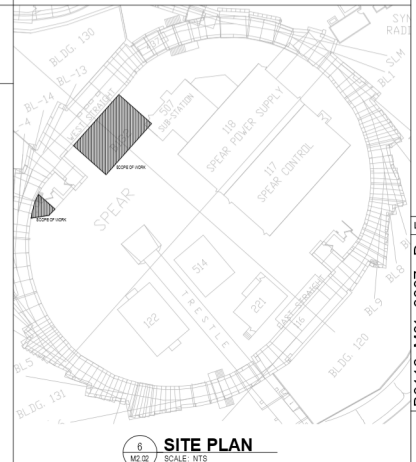
**B0132 - MECHANICAL PARKING LOT AND FLOOR PLAN**  
SCALE: 1/4" = 1'-0"

#### PLAN GENERAL NOTES

- EXISTING MECHANICAL TO REMAIN, UNLESS OTHERWISE NOTED.
- SEE ELECTRICAL DRAWINGS FOR ALL ELECTRICAL WORK.
- PROVIDE MANUAL VOLUME BALANCING DAMPERS AT EACH BRANCH TAKE-OFF IN ALL RETURN DUCTWORK. LOCATE BALANCING DAMPERS AS CLOSE TO THE BRANCH TAKE-OFF AS POSSIBLE.
- CONTRACTOR SHALL PROVIDE DUCT MATERIAL SUBMITTALS TO OWNER AND TO THE DESIGN ENGINEER FOR REVIEW AND APPROVAL, PRIOR TO PURCHASING AND INSTALLATION.
- ANY NEW OR EXISTING DUCTWORK OFFSETS REQUIRED AS A RESULT OF EXISTING JOB CONDITIONS OR LACK OF COORDINATION WITH OTHER TRADES, SHALL BE PROVIDED AT NO ADDITIONAL COST TO OWNER AND SUBJECT TO OWNER'S REVIEW.
- CONTRACTOR SHALL PROVIDE DUCTWORK AND TRANSITIONS EQUAL TO FREE AREA OF DUCTWORK AS SHOWN ON DRAWINGS TO PREVENT A CONFLICT WITH EXISTING CONDITIONS.
- ALL DUCTWORK SHALL BE SUPPORTED ACCORDING TO SMACNA AND SEISMICALLY BRACED PER SMACNA'S SEISMIC RESTRAINT MANUAL TO MEET BUILDING CODE.
- THE CONTRACTOR SHALL BALANCE ALL EQUIPMENT SHOWN ON THIS DOCUMENT PER THE FLOW QUANTITIES.
- CONTRACTOR SHALL COORDINATE WITH FINAL INSPECTION OF COMPLETED WORK WITH DESIGN ENGINEER AND OWNER OR OWNER'S REPRESENTATIVE, FOR FINAL APPROVAL AND ACCEPTANCE.
- CONTRACTOR SHALL VERIFY ALL EXISTING BUILDING CONDITIONS.

#### MECHANICAL SHEET NOTES

- PROVIDE NEW LIEBERT P4209U 8-TON UP-FLOW UNIT. INSTALL 30" x 2" FROM EXISTING LOCATION FROM THE DISCHARGE WALL THAN THE OLD FAN COIL LOCATION. MAINTAIN CODE REQUIRED CLEARANCES TO NEARBY ELECTRICAL PANEL AND EQUIPMENT. PROVIDE DUCT SMOKE DETECTOR FOR LOCAL AUTOMATIC SHUT-DOWN. FAN COIL SHALL BE LOCALLY CONTROLLED AND MAINTAIN 88°F (+/- 1°F). FIELD VERIFY FOR ANY OBSTRUCTION. PROVIDE FLOOR MOUNTING PER STRUCTURAL DRAWINGS.
- PROVIDE NEW 34X18X2 RETURN AIR PLENUM WITH FLEXIBLE CONNECTION TO RETURN AIR INLET OF FANCOIL UNIT. COMPLETE WITH DUCT INSULATION.
- PROVIDE NEW 34X10 CONDENSATE DRAIN LINE FROM FANCOIL AND EXTEND AND RE-CONNECT TO EXISTING 3/4" CONDENSATE DRAIN. TERMINATE TO HUB DRAIN AS SHOWN ON PLAN. FIELD VERIFY FOR EXACT LOCATIONS.
- PROVIDE NEW 30X18 RETURN AIR DUCT WITH 1" INSULATION.
- PROVIDE NEW 22X22 SUPPLY AIR DUCT FROM FANCOIL OUTLET AND OFFSET TO ADJACENT ROOM CEILING. PROVIDE FLEXIBLE DUCT CONNECTION AT FANCOIL OUTLET. PROVIDE DUCT INSULATION AND UNISTRUT DUCT SUPPORTS FROM CEILING STEEL BEAMS. SEE DETAILS ON M0.02.
- PROVIDE NEW 30X30 SIDEWALL RETURN AIR GRILLE AT EXISTING WALL OPENING CLOSE TO CEILING. ENLARGE OPENING FOR LARGER GRILLE SIZE.
- EXISTING CEILING LIGHTS TO REMAIN. COORDINATE WITH ELECTRICAL AND FIELD VERIFY FOR EXACT LOCATION.
- PROVIDE NEW 4-WAY ADJUSTABLE 24X24 CEILING SUPPLY AIR DIFFUSER. COMPLETE WITH OPPOSED BLADE DAMPER AND RAPID MOUNT CEILING FRAME.
- PROVIDE NEW 24X24X8 PLENUM ABOVE 24X24 CEILING SUPPLY AIR DIFFUSER AND CONNECT TO NEW SUPPLY AIR BRANCH DUCT.
- PROVIDE NEW 14X14 SUPPLY AIR BRANCH DUCT. PROVIDE DUCT INSULATION AND UNISTRUT DUCT SUPPORTS FROM CEILING STEEL BEAMS. SEE DETAILS ON M0.02.
- PROVIDE NEW LIEBERT M4206 OUTDOOR CONDENSING UNIT ON 81.5'X80.5' CONCRETE PAD. INSTALL PER MANUFACTURER'S INSTALLATION REQUIREMENTS AND CLEARANCES. EQUIPMENT IS LOCATED IN A SECURE LOCATION. COORDINATE WITH ELECTRICAL FOR POWER CONNECTION. REFER TO STRUCTURAL DRAWINGS FOR MOUNTING DETAILS.
- NEW REFRIGERANT PIPING BETWEEN NEW FAN COIL AND OUTDOOR CONDENSING UNIT. RE-USE EXISTING UNISTRUT MOUNT. PROVIDE NEW UNISTRUT MOUNT ABOVE EXISTING CONCRETE ROOF AS REQUIRED. REFRIGERANT PIPING EXPOSED TO WEATHER SHALL BE PROVIDED WITH A WATER RETARDANT COVER OVER THE INSULATION. ADHESIVE TRAP SHALL NOT BE USED FOR PROTECTION. FIELD VERIFY FOR EXACT EXISTING UNISTRUT LOCATIONS, ROUTING, AND CONDITIONS.
- NEW 6" Ø BOLLARD. COORDINATE WITH SLAC TO CONFIRM EXACT NUMBER OF BOLLARDS AND LOCATION.
- NEW UNISTRUT DUCT SUPPORT EVERY 8' FROM STEEL BEAM ABOVE. PROVIDE BEAM CLAMP, 3/8" Ø THREADED ROD, ROD STIFFENER, AND UNISTRUT SUPPORTS. SEE DETAILS ON M0.02. FIELD VERIFY EXISTING STEEL BEAM LOCATIONS PRIOR TO INSTALLATION.



**SITE PLAN**  
SCALE: NTS

|                                                                                                                                                                                                                                                                                                          |                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| SLAC BUILDING INSPECTION OFFICE                                                                                                                                                                                                                                                                          |                 |
| Construction Authorization (C)                                                                                                                                                                                                                                                                           |                 |
| BY: owner                                                                                                                                                                                                                                                                                                | DATE: 10/2/2020 |
| REVIEW: M2.02                                                                                                                                                                                                                                                                                            | DATE: 10/2/2020 |
| These plans have been reviewed and approved for construction by the SLAC Building Inspection Office. The contractor shall be responsible for obtaining all necessary permits and approvals from the appropriate authorities. THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS. |                 |



SCALE: AS NOTED DO NOT SCALE DRAWING S5010 20/25 01

|                                                                                                                                                                           |                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| SLAC U.S. DEPARTMENT OF ENERGY                                                                                                                                            |                 |
| THESE DRAWINGS, SPECIFICATIONS AND OTHER DATA HEREIN PROVIDED SHALL NOT BE COPIED, REPRODUCED OR OTHERWISE FURTHER DISSEMINATED WITHOUT PRIOR WRITTEN PERMISSION OF SLAC. |                 |
| ENGR: [Signature]                                                                                                                                                         | DATE: 10/2/2020 |
| DES: [Signature]                                                                                                                                                          | DATE: 10/2/2020 |
| CHK: [Signature]                                                                                                                                                          | DATE: 10/2/2020 |

|                                                |             |
|------------------------------------------------|-------------|
| B132 B116 AND B132 HVAC Design MECHANICAL PLAN |             |
| DRAWING NUMBER: B0116-M01-0007                 | REVISION: D |

| REV | DESCRIPTION | DRN | CHK | APP | DATE       |
|-----|-------------|-----|-----|-----|------------|
| A   | 50% CD      | HF  | JO  | SK  | 10-06-2025 |
| B   | 100% CD     | HF  | JO  | SK  | 11-06-2025 |
| C   | FAO REVIEW  | HF  | JO  | SK  | 11-24-2025 |
| D   | BIO REVIEW  | HF  | JO  | SK  | 12-18-2025 |





DEMOLITION GENERAL NOTES

1. THE QUANTITY AND LOCATION OF (E) ELECTRICAL, BUSES, PATHWAYS AND EQUIPMENT THAT SHALL BE REMOVED AS PART OF THIS PROJECT SHALL BE SHOWN ON THE REMOVAL SCHEDULE AND REMOVAL DRAWINGS BASED ON OWNER AND PARTIAL AS-BUILT DRAWINGS, AND THE CONTRACTOR SHALL NOT EXCEEDING CODES BE REMOVED WITHOUT REMOVAL DRAWING REVISED AND DISTURBING EQUIPMENT LOCATION. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING PERMITS FOR REFERENCE ONLY, AND PRIOR TO BID, SHALL SET THE SITE AND OBTAIN ALL (E) CONSTRUCTION PERMITS SHALL BE OBTAINED FOR THE RESPECTIVE DEMOLITION ACTIVITIES TO PERFORM THE DEMOLITION OF ALL ELECTRICAL SYSTEMS AND EQUIPMENT, INCLUDING ALL ELECTRICAL DEMOLITION DRAWINGS AND AS NECESSARY TO SWITCH OVER TO THE EXISTING ELECTRICAL SYSTEMS. THE CONTRACTOR SHALL OBTAIN ALL DEMOLITION DRAWINGS, WITNESS E.O.C. BY OWNER'S PERSONNEL.
2. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS FROM THE OWNER REPRESENTATIVE. ALL OTHER DISPOSERS ASSIGNED TO ELECTRICAL AND UTILITIES SHALL BE RESPONSIBLE FOR OBTAINING, BUT NOT LIMITED TO THE ELECTRICAL UTILITY PROVIDER.
3. THE CONTRACTOR SHALL COORDINATE ALL DEMOLITION WORK WITH THE ELECTRICAL UTILITY PROVIDER.
4. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY MATERIALS ADJUSTMENT REPORT PREPARED BY INDUSTRIAL HYDRAULIC ENGINEER REQUIRED FOR THE DEMOLITION OF THE ELECTRICAL SYSTEMS.
5. THE CONTRACTOR SHALL MAINTAIN AND VERIFY THE QUANTITY OF SYSTEMS REMOVED FROM ELECTRICAL SYSTEMS.
6. THE CONTRACTOR SHALL BE RESPONSIBLE FOR NOT LIMITED OF THE DEMOLITION WORK SHOWN ON THESE DEMOLITION DRAWINGS AND THE FOLLOWING:
  - A. DISCONNECT AND REMOVE (E) ELECTRICALS AND RACEWAYS MADE SPECIFIED TO DO SO.
  - B. DISCONNECT AND REMOVE (E) CONDUCTORS AND RACEWAYS MADE SPECIFIED TO DO SO.
  - C. REMOVE AND DISPOSAL OF ALL REMOVED CONDUITS, RACEWAYS AND RACEWAYS, THE PREMISES ACCORDING TO FEDERAL AND STATE REGULATIONS.
  - D. PATCH, REPAIR AND PATCH ALL SURFACES ON WALLS AND FLOORS TO BE REPAIR TO MATCH THE REMOVED SURFACES TO MATCH (E) ADJACENT CONDUITS, AND MAINTAIN (E) REPAIR PATCHED TO MATCH THE REMOVED SURFACES.
  - E. ALL (E) EQUIPMENT ITEMS REMOVED SHALL REMAIN AS THE PROPERTY OF THE OWNER AND SHALL BE PREPARED FOR STORAGE.
  - F. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY DEMOLITION AND REMOVAL DRAWINGS FOR STORAGE.
  - G. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY ELECTRICAL CONTRACTOR SHALL PREPARE TEMPORARY LIGHTING AND TEMPORARY ELECTRICAL SYSTEMS TO MAINTAIN THE PROPERTIES AS REQUIRED DURING THE CONSTRUCTION OPERATION.
  - H. THE CONTRACTOR SHALL BE RESPONSIBLE FOR OBTAINING ALL NECESSARY PERMITS TO BE PROTECTED DURING THE CONSTRUCTION.
7. ALL (E) ELECTRICAL, SIGNAL, AND FIRE ALARM EQUIPMENT, DEVICES, AND MATERIALS SHOWN ON THE REMOVAL SCHEDULE AND REMOVAL ADDITIONAL EQUIPMENT IDENTIFIED BY ORAL OR AS-BUILT OR AS-BUILT DRAWINGS SHALL BE REMOVED AND NOT LIMITED OF THE DEMOLITION DRAWING DURING THE CONSTRUCTION OPERATION UNLESS OTHERWISE INSTRUCTED BY THE OWNER'S PERSONNEL.
8. ELECTRICAL DEMOLITION SHALL COMPLY WITH THE CONSTRUCTION

### DEMOLITION KEY NOTES

- 1 SEE MECHANICAL SHEETS FOR REMOVAL OF (E) EQUIPMENT.
- 2 REMOVE (E) FEEDER BETWEEN (E) FAN COIL AND (E) DISCONNECT SWITCH.
- 3 DISCONNECT (E) CONDUCTORS AT (E) CONDENSER DISCONNECT SWITCH AND REMOVE W/ FLEX TO NEAREST UNION. PULL OUT (E) CONDUCTORS TO SOURCE PANNEL, REMOVE (E) DISCONNECT SWITCH. PREPARE (E) CONDUIT SECTION FOR RE-USE.
- 4 DISCONNECT (E) HVAC CONTROL CONDUCTORS AND REMOVE W/ FLEX TO THE NEAREST UNION. REMOVE ALL (E) CONDUCTOS BETWEEN (E) CONDENSER AND (E) FANCOIL. PREPARE (E) CONDUIT SECTION FOR RE-USE.
- 5 DISCONNECT (E) FEEDER KYTS. FROM (E) PANNEL. REMOVE CONDUIT TO THE NEAREST UNION. PREPARE ALL (E) CONDUCTORS BETWEEN (E) PANNEL TO (E) CONDENSER DISCONNECT SWITCH.



All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.



|                                                                                                                                                                                |                        |                         |                                                                       |  |                        |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------------|-----------------------------------------------------------------------|--|------------------------|
| SCALE: AS NOTED                                                                                                                                                                |                        | DO NOT SCALE DRAWING    | 55010 2025 01                                                         |  | E3.01D                 |
|      |                        |                         | B116 and B132<br>B116 and B132 HVAC DESIGN<br>ELECTRICAL<br>DEMO PLAN |  |                        |
| THE DRAWINGS, SPECIFICATIONS AND OTHER DATA HEREIN PROVIDED SHALL NOT BE REPRODUCED, COPIED, OR OTHERWISE USED OR RESEMBLED WITHOUT THE WRITTEN PERMISSION OF SLAC AND/OR DOE. |                        |                         |                                                                       |  |                        |
| ENDR. NO. 0000<br>DATE: 0000-00-00                                                                                                                                             | DATE: 00-00-00<br>0000 | APPROVALS<br>DATE: 0000 | DRAWING NUMBER<br><b>B0116-ED1-0001</b>                               |  | REVISION<br><b>D E</b> |

PLAN GENERAL NOTES

- ### PLAN KEY NOTES

- 2 B132 RF CTRL. ROOM HVAC SCALE: 1/4" = 1'-0" NORTH

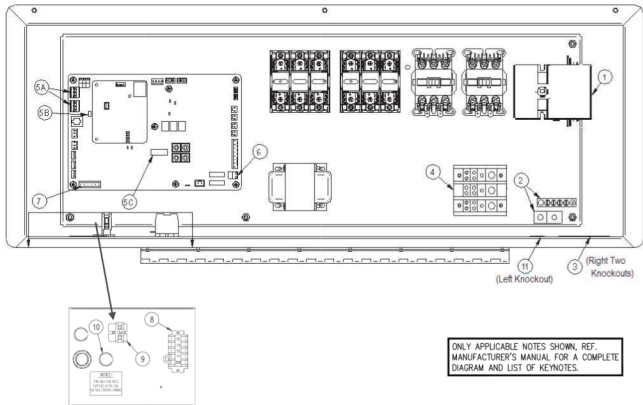
All changes from the BIO approved job set of plans and specifications shall be submitted to and approved by the BIO office prior to the changes being made in the field.

|                    |   |
|--------------------|---|
| B0116-EP1-0012 - D | E |
|--------------------|---|



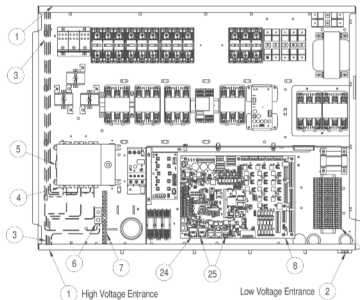


| REV | DESCRIPTION | DRN | CHK | APP | DATE       |
|-----|-------------|-----|-----|-----|------------|
| A   | 50% CD      | HF  | JO  | SK  | 10-06-2025 |
| B   | 100% CD     | HF  | JO  | SK  | 11-06-2025 |
| C   | FAO REVIEW  | HF  | JO  | SK  | 11-24-2025 |
| D   | BIO REVIEW  | HF  | JO  | SK  | 12-18-2025 |



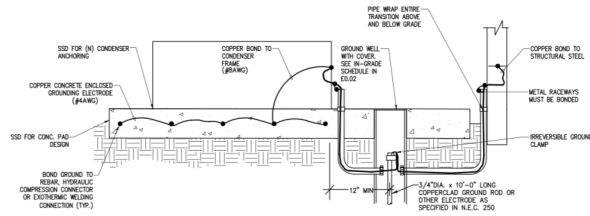
- Three phase electrical service – Terminals are on top of disconnect switch for one and two fan units. Terminals are on bottom of disconnect switch for three and four fan units. Three phase service not by Vertiv. See note 5.
- Primary high voltage entrance – Two 7/8" (22.2mm) diameter knockouts located at the bottom of the enclosure.
- Primary low voltage entrance – One 7/8" (22.2mm) diameter knockout that is free for customer low voltage wiring.

7 LIEBERT MC ELECTRICAL WIRING DIAGRAM  
CONDENSER UNIT

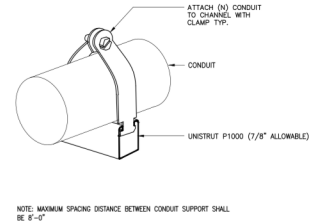


- High Voltage Entrance. Supplied on top and bottom of electric box. Knockout size Ø1.75in (44.5mm).
- Low Voltage Entrance. Ø1.375 in. (34.9mm) hole located on bottom of Electric Box.
- Three phase Electric Service and earth ground. Field supplied.
- Three phase connection. Electric service connection terminals on disconnect.
- Factory installed disconnect switch. Fused disconnect switch provided on units.
- Earth ground connection. Connection terminals for field supplied earth grounding wire.
- Earth ground bar. Connection terminals with factory ground from each high voltage component for field supplied earth grounding wire.

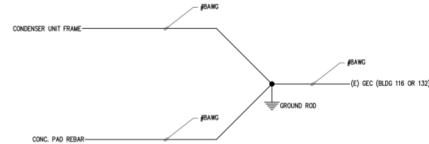
8 LIEBERT PDX ELECTRICAL WIRING DIAGRAM  
FAN COIL UNIT



4 PAD MOUNTED  
EQUIPMENT BONDING

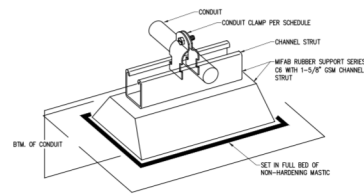


1 TYP. CONDUIT SUPPORT DETAIL  
CHANNEL STRUT

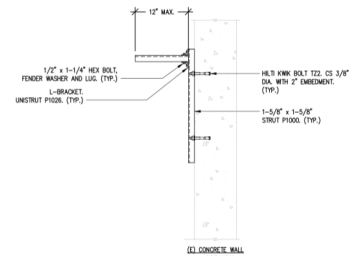


5 CONDENSER AND CONC. PAD REBAR GROUNDING

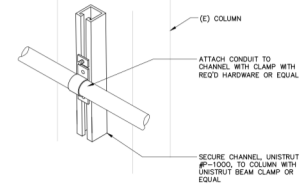
|            | SUPPORT OF CONDUIT |           |                                  |
|------------|--------------------|-----------|----------------------------------|
|            | STD PIPE SIZE      | CLAMP     | CLAMP SPACING (CENTER TO CENTER) |
| HORIZONTAL | 1/2 (0.840 O.D.)   | P1111     | 5'-0"                            |
|            | 3/4 (1.050 O.D.)   | P1112     | 6'-0"                            |
|            | 1 (1.315 O.D.)     | P1113     | 8'-0"                            |
|            | 1 1/4 (1.660 O.D.) | P1114     | 10'-0"                           |
|            | 1 1/2 (1.900 O.D.) | P1115     | 12'-0"                           |
|            | 2 (2.375 O.D.)     | P1117     | 16'-0"                           |
| VERTICAL   | SEE ABOVE          | SEE ABOVE | EVERY FLOOR LEVEL                |



6 TYP. CONDUIT SLEEPER SUPPORT  
ROOF SURFACE



2 TYP. CHANNEL STRUT SUPPORT DETAIL  
CONC. WALL



3 TYP. CONDUIT SUPPORTS  
COLUMNS OR WALLS

|                                                                                                                                                                                                                                                                                                                      |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| SLAC BUILDING INSPECTION OFFICE                                                                                                                                                                                                                                                                                      |             |
| Construction Authorization (FC)                                                                                                                                                                                                                                                                                      |             |
| BY: <input type="checkbox"/> order                                                                                                                                                                                                                                                                                   |             |
| DATE: 10/2/2025                                                                                                                                                                                                                                                                                                      | PRJ# 25-204 |
| Revised Authorization: <input checked="" type="checkbox"/> No <input type="checkbox"/> Yes                                                                                                                                                                                                                           |             |
| These plans have been reviewed and approved for construction. The contractor is responsible for obtaining all necessary permits and for compliance with all applicable codes and regulations. The authorization of these plans does not constitute a warranty or endorsement of any products, materials, or methods. |             |
| THESE PLANS SHALL BE KEPT ON THE JOB SITE FOR ALL CONSTRUCTION INSPECTIONS.                                                                                                                                                                                                                                          |             |

All changes from the BIO approved job set of plans and specifications shall be submitted to, and approved by the BIO office prior to the changes being made in the field.

**vektor** Engineering & Consulting Services, Inc.  
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San Ramon, CA 94583  
+1 (925) VEKTOR1 (925-8871)



|                                                                                                                                                                                              |                                                                   |                                                                        |                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------|
| SCALE: AS NOTED                                                                                                                                                                              | DO NOT SCALE DRAWING                                              | S5010 2025.01                                                          | E6.01                                         |
| <b>SLAC</b> U.S. DEPARTMENT OF ENERGY<br>NATIONAL ACCELERATOR LABORATORY                                                                                                                     |                                                                   | B116 AND B132<br>B116 AND B132 HVAC DESIGN<br>ELECTRICAL<br>DETAILS    |                                               |
| THE DRAWING SPECIFICATIONS AND OTHER DATA HEREIN PROVIDED SHALL NOT BE COPIED, REPRODUCED OR OTHERWISE IN ANY MANNER WITHOUT THE WRITTEN PERMISSION OF SLAC NATIONAL ACCELERATOR LABORATORY. |                                                                   |                                                                        |                                               |
| ENGR: HENRI J. BERRY<br>DES: PETER J. HARRIS<br>CHK: JEFFREY L. HARRIS                                                                                                                       | DATE: 10/2/2025<br>DES: PETER J. HARRIS<br>CHK: JEFFREY L. HARRIS | APPROVALS: 10/2/2025<br>DES: PETER J. HARRIS<br>CHK: JEFFREY L. HARRIS | DRAWING NUMBER: B0116-E05-0015<br>REVISION: D |

